Home Work 1

Q1. Add Two Integers

Write a C++ program that takes two integers as input and displays their sum.

Q2. Subtract Two Integers

Write a C++ program that takes two integers and calculates their difference.

Q3. Multiply Two Integers

Write a C++ program that takes two integers and displays their product.

Q4. Divide Two Numbers

Write a C++ program that takes two integers and displays their division result using `float`.

Q5. Calculate the Remainder

Write a C++ program that takes two integers and displays the remainder using the `%` operator.

Q6. Calculate the Average

Write a C++ program that takes three integer marks and calculates their average using `float`.

Q7. Calculate the Square

Write a C++ program that takes an integer and calculates its square.

Q8. Calculate the Cube

Write a C++ program that takes an integer and calculates its cube.

Q9. Area of a Rectangle

Write a C++ program that takes the length and width of a rectangle and calculates its area.

Formula:
`Area = Length × Width`

Q10. Perimeter of a Rectangle

Write a C++ program that takes the length and width of a rectangle and calculates its perimeter.

Formula:
`Perimeter = 2 × (Length + Width)`

Q11. Area of a Circle

Write a C++ program that takes the radius of a circle as a `float` and calculates its area.

Formula:
`Area = 3.14 × Radius × Radius`

Q12. Circumference of a Circle

Write a C++ program that takes the radius of a circle and calculates its circumference.

Formula:
`Circumference = 2 × 3.14 × Radius`

Q13. Area of a Triangle

Write a C++ program that takes the base and height of a triangle and calculates its area.

Formula:
`Area = 0.5 × Base × Height`

Q14. Celsius to Fahrenheit

Write a C++ program that takes temperature in Celsius and converts it to Fahrenheit.

Formula:
`F = (C × 9 / 5) + 32`

Q15. Fahrenheit to Celsius

Write a C++ program that takes temperature in Fahrenheit and converts it to Celsius.

Formula:
`C = (F - 32) × 5 / 9`

Q16. Simple Interest

Write a C++ program that takes principal, rate, and time and calculates simple interest.

Formula:
`SI = (P × R × T) / 100`

Q17. Total and Average Marks

Write a C++ program that takes the marks of five subjects and calculates the total and average marks.

Q18. Percentage Calculator

Write a C++ program that takes obtained marks and total marks and calculates the percentage.

Formula:

```
Percentage = (Obtained / Total) × 100
```

Q19. Calculate Profit

Write a C++ program that takes cost price and selling price and calculates the profit.

Formula:

```
Profit = Selling Price - Cost Price
```

Q20. Calculate Loss

Write a C++ program that takes cost price and selling price and calculates the loss.

Formula:

```
Loss = Cost Price - Selling Price
```

Q21. Profit Percentage

Write a C++ program that takes cost price and selling price and calculates the profit percentage.

Formula:

```
Profit = SP - CP
Profit % = (Profit / CP) × 100
```

Q22. Discount Calculation

Write a C++ program that takes the original price and discount percentage. Calculate the discount amount and final price.

Q23. Salary Calculation

Write a C++ program that takes basic salary, house allowance, and transport allowance and calculates the total salary.

Q24. Electricity Bill

Write a C++ program that takes electricity units consumed and cost per unit and calculates the total bill.

Q25. Shopping Bill

A customer purchases three products. Input the price and quantity of each product and calculate the total bill.

Q26. BMI Calculator

Write a C++ program that takes weight in kilograms and height in meters and calculates BMI.

Formula:

```
BMI = Weight / (Height × Height)
```

Q27. Time Conversion

Write a C++ program that takes total seconds and converts them into hours, minutes, and seconds.

Q28. Currency Conversion

Write a C++ program that takes an amount and an exchange rate and calculates the converted amount.

Formula:

```
Converted Amount = Amount × Exchange Rate
```

Q29. Daily Expenses

Write a C++ program that takes five daily expenses and calculates the total and average expense.

Q30. Student Result

Write a C++ program that takes marks of six subjects and calculates total marks, average marks, and percentage.

Q31. Three-Number Calculation

Input three integers `A`, `B`, and `C`.

Calculate:

```
Result = (A + B) × C
```

Then calculate:

```
Final = Result / (A + C)
```

Display both results.

Q32. Two-Digit Number Operations

Input a two-digit integer. Extract the tens and ones digits.

Calculate:

- Sum of digits
- Product of digits
- Difference between digits

Q33. Three-Digit Number Operations

Input a three-digit integer. Extract the hundreds, tens, and ones digits. Calculate their sum and product.

Q34. Swap Two Numbers

Write a C++ program that swaps two integer values using a third variable.

Q35. Swap Without a Third Variable

Write a C++ program that swaps two integer values without using a third variable.

Q36. Mathematical Expression

Input `A`, `B`, and `C` and calculate:

```
Result = (A² + B² + C²) / (A + B + C)
```

Use `float` for the result.

Q37. Weighted Average

A student's marks are weighted as follows:

- Assignment = 20%
- Midterm = 30%
- Final Exam = 50%

Input all three marks and calculate the final weighted score.

Q38. Gross Salary

Input basic salary and the percentage of:

- House allowance
- Medical allowance
- Transport allowance

Calculate each allowance and gross salary.

Q39. Net Salary

Input basic salary, allowance, and tax percentage.

Calculate:

```
Gross Salary = Basic Salary + Allowance

Tax = Gross Salary × Tax% / 100

Net Salary = Gross Salary - Tax
```

Q40. Shopping Bill with Tax

Input the price and quantity of three products.

Calculate:

- Subtotal
- 17% tax
- Final bill

Q41. Discount and Tax

Input price, discount percentage, and tax percentage.

Calculate:

- Discount
- Price after discount
- Tax
- Final price

Q42. Car Trip Cost

Input distance, car mileage, and fuel price per liter.

Calculate:

```
Fuel Used = Distance / Mileage

Fuel Cost = Fuel Used × Fuel Price
```

Q43. Electricity Bill with Tax

Input electricity units, price per unit, and service charge.

Calculate the energy cost, total bill, and final bill after adding 5% tax.

Q44. Loan Interest

Input loan amount, annual interest rate, and number of months.

Calculate the interest and total amount to be paid.

Q45. Rectangle Diagonal

Input the length and width of a rectangle and calculate its diagonal.

Formula:

```
Diagonal = √(Length² + Width²)
```

Q46. Distance Between Two Points

Input `x1`, `y1`, `x2`, and `y2`.

Calculate the distance between the two points.

Formula:

```
Distance = √((x2-x1)² + (y2-y1)²)
```

Q47. Average Speed

A person travels two distances at two different speeds.

Input:

- Distance 1
- Speed 1
- Distance 2
- Speed 2

Calculate total distance, total time, and average speed.

Q48. Temperature Conversion

Input Celsius and calculate both Fahrenheit and Kelvin.

Formulas:

```
F = (C × 9 / 5) + 32

K = C + 273.15
```

Q49. Age Conversion

Input age in years and calculate approximate age in:

- Months
- Weeks
- Days
- Hours

Use:

```
1 year = 12 months
1 year = 52 weeks
1 year = 365 days
1 day = 24 hours
```

Q50. Time Conversion

Input total seconds and convert them into hours, minutes, and seconds.

Q51. Restaurant Bill

Input food, drink, and dessert prices with their quantities.

Calculate:

- Subtotal
- 10% tax
- 15% tip
- Final bill

Q52. Student Result Analysis

Input marks of six subjects.

Calculate:

- Total
- Average

- Percentage
- Marks required to reach 80%

Q53. Exam Marks Analysis

Input marks of four exams.

Calculate:

- Total marks
- Average
- Percentage
- Difference between the highest and lowest marks

Q54. Bank Account Calculation

Input:

- Initial balance
- Deposit
- Withdrawal
- Interest rate

Calculate the balance, interest, and final balance.

Q55. Compound Interest

Input principal, annual interest rate, and number of years.

Calculate compound interest.

Formula:

```
Amount = P × (1 + R/100)^T

Compound Interest = Amount - P
```

Q56. Investment Return

Input initial investment, annual return percentage, and number of years.

Calculate the final investment value using compound growth.

Q57. BMI and Target Weight

Input weight and height.

Calculate BMI and the weight required to achieve a BMI of 22.

Formula:

```
BMI = Weight / Height²

Target Weight = 22 × Height²
```

Q58. Manufacturing Cost

Input:

- Raw material cost
- Labor cost
- Machine cost
- Packaging cost
- Number of products

Calculate total manufacturing cost and cost per product.

Q59. Travel Expense Calculator

Input:

- Total distance
- Fuel efficiency
- Fuel price
- Hotel cost
- Food cost per day
- Number of days

Calculate fuel cost, food cost, and total travel expense.

Q60. Complete Salary Calculator

Input:

- Basic salary
- House allowance %
- Medical allowance %
- Transport allowance %
- Tax %
- Insurance %

Calculate:

- Each allowance

- Gross salary
- Tax
- Insurance
- Net salary